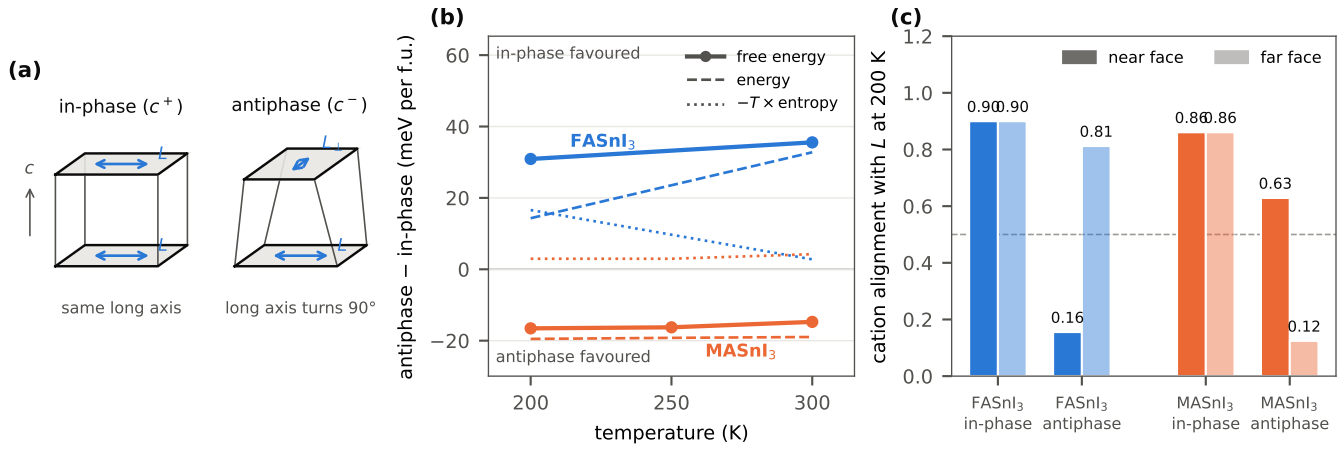
(a)
in-phase (c^+)
antiphase (c^-)
c
L
same long axis
long axis turns 90°
(b)
in-phase favoured
antiphase favoured
free energy
energy
$-T \times$ entropy
$FASnI_3$
$MASnI_3$
antiphase − in-phase (meV per f.u.)
temperature (K)
(c)
near face
far face
cation alignment with L at 200 K
0.90
0.90
0.16
0.81
0.86
0.86
0.63
0.12
$FASnI_3$ in-phase
$FASnI_3$ antiphase
$MASnI_3$ in-phase
$MASnI_3$ antiphase